Global Hilbert expansion for some non-relativistic kinetic equations

Qinghua Xiao 肖清华

Innovation Academy for Precision Measurement Science and Technology, CAS

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Hilbert expansion (1912)

In 1912, Hilbert proposed the Hilbert expansion (the first result in Hilbert's sixth problem):

Boltzmann equation $\Rightarrow$ compressible Euler equations.

Namely,

$$\begin{cases} \partial_t F^\varepsilon + v \cdot \nabla_x F^\varepsilon = \frac{1}{\varepsilon} Q(F^\varepsilon, F^\varepsilon), \\ F^\varepsilon = \sum_{n=0}^{\infty} \varepsilon^n F_n, \quad (\varepsilon : \text{the Knudsen number}). \end{cases} \Rightarrow$$

$$\begin{cases} \varepsilon^{-1} : \quad Q(F_0, F_0) = 0, \\ \varepsilon^0 : \quad \partial_t F_0 + v \cdot \nabla_x F_0 = Q(F_0, F_1) + Q(F_1, F_0). \end{cases} \Rightarrow$$

$$\begin{cases} F_0 := \mathbf{M} = \frac{\rho}{(2\pi T)^{3/2}} \exp\left\{ - \frac{|v-u|^2}{2T} \right\}, \\ (\rho, u, T) \text{ satisfies the compressible Euler equations.} \end{cases}$$

Breakthrough by Caflisch

The first breakthrough was made by Caflisch (CPAM-1980) with a truncated Hilbert expansion:

$$F^{\varepsilon} = \sum_{n=0}^{2k-1} \varepsilon^n F_n + \varepsilon^k F_R^{\varepsilon}, \quad k \geq 2.$$

♦ Coefficients $F_n (0 \leq n \leq 2k-2)$: can be solved successively based on (ρ, u, T).

♦ Main difficulty in solving the remainder term $F_R^{\varepsilon} = \mathbf{M}^{\frac{1}{2}} f^{\varepsilon}$:

$$f^{\varepsilon}(t,x,v)\mathbf{M}^{-\frac{1}{2}}(t,x,v)\Big\{\partial_t + v\cdot\nabla_x\Big\}\mathbf{M}^{\frac{1}{2}}(t,x,v) \sim \langle v\rangle^3 f^{\varepsilon}(t,x,v) \quad (1)$$

Breakthrough by Caflisch

♦ Via Caflisch's decomposition :

$$F_R^\varepsilon = \underbrace{\mathbf{M}^{\frac{1}{2}} f^\varepsilon}_{\text{low velocity part}} + \underbrace{\mu^{\frac{1}{2}} h^\varepsilon}_{\text{high velocity part}},$$

where

$$\mu(v) = \frac{1}{(2\pi T_c)^{3/2}} \exp\left\{-\frac{|v|^2}{2T_c}\right\}, \quad \frac{1}{2}T(t,x) < T_c < T(t,x),$$

F_R^ε was solved as a system

$$\begin{cases} \big(\partial_t + v\cdot\nabla_x\big) f^\varepsilon = F(f^\varepsilon, h^\varepsilon), \\ \big(\partial_t + v\cdot\nabla_x\big) h^\varepsilon = G(f^\varepsilon, h^\varepsilon), \\ f^\varepsilon(0,x,v) = h^\varepsilon(0,x,v) = 0. \end{cases}$$

♦ Drawback: $F^\varepsilon(t,x,v) \geq 0$ can not be guaranteed.

Methods to study the Hilbert expansion

L^2–L^∞ method (inspired by Guo-ARMA-2010):
first introduced in Guo-Jang-Jiang (KRM-2009, CPAM-2010).

♦ Main ideas: interplay estimates of L^2-norm and L^∞-norm.

♦ Key estimates:

$$\left\langle \left| \mathbf{M}^{-\frac{1}{2}} (\partial_t + v \cdot \nabla_x) \mathbf{M}^{\frac{1}{2}} \right| f^\varepsilon, f^\varepsilon \right\rangle \lesssim \|\nabla_{x,t}(\rho, u, T)\| \, \|h^\varepsilon\|_\infty \|f^\varepsilon\|,$$

$$\|h^\varepsilon\|_\infty \lesssim \varepsilon^{\frac{3}{2}} \|f^\varepsilon\| + \ldots\ldots,$$

where $\beta > \frac{9-2\gamma}{2}$ and h^ε satisfies

$$(1 + |v|)^{-\beta} \mu^{\frac{1}{2}} h^\varepsilon = \mathbf{M}^{\frac{1}{2}} f^\varepsilon = F_R^\varepsilon, \quad \frac{1}{2} T(t,x) < T_c < T(t,x).$$

♦ Further developments:

- VPB system: Guo-Jang-CMP-2010
- r-Boltzmann equation: Speck-Strain-CMP-2011
- r-VMB system: Guo-Xiao-CMP-2021

Remark: Since the L^2–L^∞ method heavily relies on the characteristic method, it seems difficult to be applied in the Landau type and non-cutoff Boltzmann type equations.

Problem: What about the Hilbert expansion of Landau type and non-cutoff Boltzmann type equations?

Methods to study the Hilbert expansion

Exponentially weighted energy method (inspired by Duan-Yang-Zhao (JDE-2012, M3AS-2013)):

- first introduced in Ouyang-Wu-Xiao (arXiv:2207.00126)
- further development in Lei-Liu-Xiao-Zhao (arXiv:2209.15201, arXiv:2310.11745)

♦ Main ideas:

- overcome the velocity growth via a special weight function
- interplay energy estimates involving local and global Maxwellians

♦ Key estimates: For $w_e = \exp\left\{\frac{\bar{c}\langle v\rangle^a}{\ln(\mathrm{e}+t)T_c}\right\}, \bar{c} > 0, 1 \le a \le 2,$

$F_R^\varepsilon = \mathbf{M}^{\frac{1}{2}} f^\varepsilon = \mu^{\frac{1}{2}} h^\varepsilon,$

$$w_e^2 f^\varepsilon \partial_t f^\varepsilon = \frac{1}{2}\frac{\partial}{\partial t}\left(w_e f^\varepsilon\right)^2 + \frac{\bar{c}\langle v\rangle^a}{(\mathrm{e}+t)[\ln(\mathrm{e}+t)]^2}(w_e f^\varepsilon)^2,$$

$$\left\|\langle v\rangle^{3/2} f^\varepsilon\right\|^2 \lesssim \|f^\varepsilon\|^2 + \varepsilon^{-1}\left\|(\mathbf{I}-\mathbf{P_M})\,[f^\varepsilon]\right\|_{\mathbf{D}}^2 + e^{-\frac{C}{\sqrt{\varepsilon}}}\|h^\varepsilon\|^2 .$$

Concerned systems

Consider the scaled Vlasov-Maxwell Landau system or non-cutoff Vlasov-Maxwell-Boltzmann system:

$$\begin{cases} \partial_t F^\varepsilon + v \cdot \nabla_x F^\varepsilon - e\Big(E^\epsilon + v \times B^\varepsilon\Big) \cdot \nabla_v F^\varepsilon = \dfrac{1}{\varepsilon}\mathcal{C}\left[F^\varepsilon, F^\varepsilon\right], \\ \partial_t E^\varepsilon - c\nabla_x \times B^\varepsilon = 4\pi e \displaystyle\int_{\mathbb{R}^3} vF^\varepsilon \mathrm{d}v, \\ \partial_t B^\varepsilon + c\nabla_x \times E^\varepsilon = 0, \\ \nabla_x \cdot E^\varepsilon = 4\pi e\Big(1 - \displaystyle\int_{\mathbb{R}^3} F^\varepsilon \mathrm{d}v\Big), \\ \nabla_x \cdot B^\varepsilon = 0. \end{cases}$$

- $F^\varepsilon(t, x, v)$: distribution function for electrons
- $-e$: electrons' charge
- $\mathcal{C}\left[\cdot, \cdot\right]$: Landau (Boltzmann) collision operator
- E^ε, B^ε: electromagnetic field

Corresponding to hydrodynamic level model, Euler-Maxwell system:

$$\begin{cases} \partial_t \rho + \nabla_x \cdot (\rho u) = 0, \\ \partial_t [\rho u] + \nabla_x \cdot (\rho u \otimes u) + \frac{2}{3} \nabla_x \rho^{\frac{5}{3}} + \rho \Big(E + u \times B \Big) = \mathbf{0}, \end{cases}$$

$$\begin{cases} \partial_t E - c \nabla_x \times B = 4\pi e n u, \\ \partial_t B + c \nabla_x \times E = \mathbf{0}, \\ \nabla_x \cdot E = 4\pi e (1 - n), \qquad \nabla_x \cdot B = 0. \end{cases}$$

Theorem(Ionescu-Pausader-JEMS-2014):

Let $\Big(\rho(0,x), u(0,x), T(0,x), E(0,x), B(0,x)\Big)$ be irrotational and be close to constant state $\big(1, \mathbf{0}, 1, \mathbf{0}, \mathbf{0}\big)$. Then $\exists!$ global solution $\Big(\rho(t,x), u(t,x), T(t,x), E(t,x), B(t,x)\Big)$ to the EM system that is irrotational for any $t > 0$ and it satifies

$$\sup_{t\in[0,\infty)} \left\|(\rho(t)-1, u(t), T(t)-1, E(t), B(t))\right\|_{H^{N_0}} \tag{2}$$
$$+ \sup_{t\in[0,\infty)} \sup_{|\rho|\leq \bar{N}} \Big((1+t)^{p_0} \left\|\nabla_x^\rho (\rho(t)-1, u(t), T(t)-1, E(t), B(t))\right\|_{L^\infty}\Big) \lesssim \bar{\varepsilon}_0,$$

where $N_0 \in \mathbb{N}$ is sufficiently large, $3 \leq N_1 \ll N_0$, $p_0 = \frac{101}{100}$ and $\bar{\varepsilon}_0$ is sufficiently small.

Reference review

♦ Time-asymptotic equivalence of kinetic equations and fluid equations:

- Boltzmann type equations
- Landau type equations

♦ Parameter (Knudsen number, Strouhal number) limits of kinetic equations:

- Compressible Euler limit
- Incompressible Euler limit
- Incompressible Navier-Stokes-Fourier limit
- Compressible Navier-Stokes approximation
- Hilbert expansion

Bardos, Caflish, Duan, Golse, Guo, Huang, Jang, Jiang, Levermore, Liu, Masmoudi, Nishida, Saint-Raymond, Strain, Ukai, Wang, Yang, Yu, Zhao

……

Hilbert expansion of the VML system

Theorem (VML): Let $F^\varepsilon(0,x,p) \geq 0$. Assume that $\big(n(t,x), u(t,x), T(t,x), E(t,x), B(t,x)\big)$ is the global solution constructed in Ionescu-Pausader-JEMS-2014. Then for $k \geq 3$ in the Hilbert expansion

$$F^\varepsilon = \sum_{n=0}^{2k-1} \varepsilon^n F_n + \varepsilon^k F_R^\varepsilon, \quad E^\varepsilon = \sum_{n=0}^{2k-1} \varepsilon^n E_n + \varepsilon^k E_R^\varepsilon, \quad B^\varepsilon = \sum_{n=0}^{2k-1} \varepsilon^n B_n + \varepsilon^k B_R^\varepsilon,$$

and $f^\varepsilon = \mathbf{M}^{-\frac{1}{2}} F_R^\varepsilon$, there exists an $\varepsilon_1 > 0$ such that for $0 \leq \varepsilon \leq \varepsilon_1$ and $0 < t \leq \bar{t}$ with $\bar{t} = \varepsilon^{-1/3}$, if $\frac{\epsilon_1}{C_0} \leq T(t,x) - T_c \leq C_0 \epsilon_1$,

$$\bar{\mathcal{E}}(0) \lesssim 1,$$

The VML system admits a unique solution satisfying $F^\varepsilon(t,x,p) \geq 0$ and

$$\sup_{0 \leq t \leq \bar{t}} \bar{\mathcal{E}}(t) + \int_0^{\bar{t}} \bar{\mathcal{D}}(s) \mathrm{d}s \lesssim \bar{\mathcal{E}}(0) + 1. \tag{3}$$

Hilbert expansion of the VML system (continued)

where the energy functional $\bar{\mathcal{E}}(t)$ and dissipation rate functional $\bar{\mathcal{D}}(t)$ are defined as follows:

$$\bar{\mathcal{E}}(t) = \sum_{i=0}^{2} \varepsilon^i \Big[\Big(\Big\| \sqrt{4\pi RT} \nabla_x^i f^\varepsilon(t) \Big\|^2 + \big\| \nabla_x^i E_R^\varepsilon(t) \big\|^2 + \big\| \nabla_x^i B_R^\varepsilon(t) \big\|^2 \Big) + \varepsilon^{\frac{4}{3}} \big\| w_i \nabla_x^i h^\varepsilon(t) \big\|^2 \Big],$$

$$\bar{\mathcal{D}}(t) \backsimeq \sum_{i=0}^{2} \varepsilon^i \left[\frac{1}{\varepsilon} \left\| \nabla_x^i (\mathbf{I} - \mathbf{P_M}) \left[f^\varepsilon \right] (t) \right\|_{\mathbf{D}}^2 + \varepsilon^{\frac{1}{3}} \left\| w_i \nabla_x^i h^\varepsilon(t) \right\|_{\mathbf{D}}^2 + \varepsilon^{\frac{4}{3}} Y(t) \left\| (1+|v|) w_i \nabla_x^i h^\varepsilon(t) \right\|^2 \right],$$

where $f^\varepsilon = \mathbf{M}^{-\frac{1}{2}} F_R^\varepsilon, h^\varepsilon = \mu^{-\frac{1}{2}} F_R^\varepsilon$.

Remark:

- In the energy functional $\bar{\mathcal{E}}(t)$, compared with norm of $\nabla_x^i f^\varepsilon$, a coefficient $\varepsilon^{\frac{4}{3}}$ is added for the norm of $\nabla_x^i h^\varepsilon$ to remove the $\frac{1}{\varepsilon}$ singularity and time growth for $t \in [0, \varepsilon^{-\frac{1}{3}}]$.
- No macroscopic dissipation of f^ε or h^ε is included in $\bar{\mathcal{D}}(t)$.

Main ingredients for the VML system

Key point: an interplay of the energy estimates in the local and global Maxwellian frameworks.

(i) Estimate of the velocity growth term (1):

Let $F_R^\varepsilon = \mathbf{M}^{\frac{1}{2}} f^\varepsilon = \mu^{\frac{1}{2}} h^\varepsilon$, we have

$$|\partial^\alpha f^\varepsilon| \lesssim \sum_{j=0}^{|\alpha|} \langle v\rangle^{2(|\alpha|-j)} \exp\left(-\frac{(T-T_c)|v|^2}{8RTT_c}\right)\left|\nabla_x^j h^\varepsilon\right|, \quad 0 \le |\alpha| \le 2.$$

Take $|\alpha| = 0$ and use $\frac{\epsilon_1}{C_0} \le T(t,x) - T_c \le C_0\epsilon_1$ to have

$$\begin{aligned}
&\left\|\langle v\rangle^{3/2} f^\varepsilon\right\|^2 \\
\lesssim &\left\|\langle v\rangle^{3/2}\mathbf{P}_\mathbf{M}[f^\varepsilon]\right\|^2 + \left\|\langle v\rangle^{3/2}\left(\mathbf{I}-\mathbf{P}_\mathbf{M}\right)[f^\varepsilon]\right\|^2 \\
\lesssim &\|f^\varepsilon\|^2 + \int_{\mathbb{R}^3}\int_{\langle v\rangle^4 \le \varepsilon^{-1}} \langle v\rangle^3 \left|\left(\mathbf{I}-\mathbf{P}_\mathbf{M}\right)[f^\varepsilon]\right|^2 dxdv + \int_{\mathbb{R}^3}\int_{\langle v\rangle^4 \ge \varepsilon^{-1}} \langle v\rangle^3 |f^\varepsilon|^2 dxdv \\
\lesssim &\|f^\varepsilon\|^2 + \varepsilon^{-1}\left\|\left(\mathbf{I}-\mathbf{P}_\mathbf{M}\right)[f^\varepsilon]\right\|_\mathbf{D}^2 + C(\epsilon_1) e^{-\frac{\epsilon_1}{8C_0 R T_c^2 \sqrt{\varepsilon}}} \|h^\varepsilon\|^2.
\end{aligned}$$

(ii) Estimate of the term $\frac{1}{\varepsilon}\mathcal{L}_{\mathbf{D}}[h^\varepsilon]$:

Note that

$$-\mathcal{L}_{\mathbf{D}}[h^\varepsilon] = \mu^{-\frac{1}{2}}\left[C\left(\mathbf{M}-\mu, \mu^{\frac{1}{2}}h^\varepsilon\right) + C\left(\mu^{\frac{1}{2}}h^\varepsilon, \mathbf{M}-\mu\right)\right].$$

Since $\mathbf{M}$ and μ are sufficiently close, we have

$$\frac{1}{\varepsilon}\left|\left\langle \mathcal{L}_{\mathbf{D}}[h^\varepsilon], h^\varepsilon\right\rangle\right| \lesssim \frac{\epsilon_1}{\varepsilon}\|h^\varepsilon\|_{\mathbf{D}}^2 \lesssim \frac{\epsilon_1}{\varepsilon}\left(\|(\mathbf{I}-\mathbf{P})[h^\varepsilon]\|_{\mathbf{D}}^2 + \|\mathbf{P}[h^\varepsilon]\|_{\mathbf{D}}^2\right).$$

On the other hand, we have

$$\frac{\epsilon_1}{\varepsilon}\left\|\mathbf{P}\left[h^\varepsilon\right]\right\|_{\mathbf{D}}^2 \lesssim \frac{\epsilon_1}{\varepsilon}\left\|\mathbf{P}_{\mathbf{M}}\left[f^\varepsilon\right]\right\|^2 \lesssim \frac{\epsilon_1}{\varepsilon}\|f^\varepsilon\|^2.$$

To make the red term time-integrable over $[0, \varepsilon^{-1/3}]$, we add additional weight $\varepsilon^{4/3}$ to the norm $\|h^\varepsilon\|^2$.

Main ingredients for the VML system

(iii) Design of ε-dependent norms in $\mathcal{E}(t)$:

$$
\begin{aligned}
\left\langle \frac{1}{\varepsilon}\nabla_x \mathcal{L}[f^\varepsilon], \nabla_x f^\varepsilon \right\rangle =& \left\langle \frac{1}{\varepsilon}\mathcal{L}\Big[\nabla_x(\mathbf{I}-\mathbf{P})[f^\varepsilon]\Big], (\mathbf{I}-\mathbf{P})\nabla_x[f^\varepsilon] \right\rangle \\
&- \left\langle \frac{1}{\varepsilon}\left[\!\left[\mathcal{L}, \nabla_x\right]\!\right](\mathbf{I}-\mathbf{P})[f^\varepsilon], \nabla_x f^\varepsilon \right\rangle \\
\geq& \frac{\delta}{\varepsilon}\left\|\nabla_x(\mathbf{I}-\mathbf{P})[f^\varepsilon]\right\|_\sigma^2 - \frac{C\mathcal{Z}}{\varepsilon^2}\left\|(\mathbf{I}-\mathbf{P})[f^\varepsilon]\right\|_\sigma^2 - C\mathcal{Z}\,\|f^\varepsilon\|_{H^1}^2,
\end{aligned}
$$

where $\mathcal{Z} = \sup_{x\in\mathbb{R}^3, 0\le\ell\le 2}\left\{\left|\nabla_{t,x}^{1+\ell}(n_0, u, T)\right| + \left|\nabla_{t,x}^{\ell}(E_0, B_0)\right| + \left|\nabla_{t,x}^{\ell}u\right|\right\}$.
To control the red terms, we design $\varepsilon\left\|\nabla_x f^\varepsilon\right\|^2$ in $\mathcal{E}(t)$. Similarly, $\varepsilon^2\left\|\nabla_x^2 f^\varepsilon\right\|^2$ is also designed for the estimate of $\nabla_x^2 f^\varepsilon$.

Main ingredients for the VML system

(iv) time-dependent exponential weight function:

To proceed energy estimates of h^ε, introduce a time-related exponential function

$$w_{|\alpha|}(t,v) := \langle v\rangle^{\ell-|\alpha|}\exp\left(\frac{(1+|v|^2)}{8RT_c\ln(\mathrm{e}+t)}\right), \quad 0\le|\alpha|\le s<\ell,$$

where $|\alpha|$ is the order of $x-$derivatives.

- The exponential part is mainly designed to control velocity growth term such as $\frac{E\cdot v}{2RT_c}h^\varepsilon$
- The algebraic part is mainly designed to bound terms with $v-$derivatives like $(E+v\times B)\cdot\nabla_v h^\varepsilon$.

Remark: The weight $w_{|\alpha|}$ also leads to velocity growth in estimating $\varepsilon^{k-1}\Gamma(h^\varepsilon,h^\varepsilon)$. This problem is solved by the inequality:

$$\sup_{v\in\mathbb{R}^3}\left[\langle v\rangle^i\exp\left(\frac{-\epsilon_1\langle v\rangle^2}{8C_0RT_c\ln(\mathrm{e}+t)}\right)\right]\lesssim\left(\frac{\ln(\mathrm{e}+t)}{\epsilon_1}\right)^{\frac{i}{2}}.$$

Additional difficulties compared to the VML system:

- weak dissipation of $\mathcal{L}_{\mathbf{M}}[f]$ or $(\mathcal{L}[f])$: absence of $\nabla_v f$; $\gamma + 2 \to -1$ for the Landau case and $\gamma + 2s \to -\frac{3}{2}$ for the non-cutoff Boltzmann case
- no estimate with weight $e^{\frac{q}{2}\langle v \rangle^2}(0 < q < 1)$ available (only $e^{\frac{q}{2}\langle v \rangle}(0 < q < \infty)$ type weight doable)

Main ingredients for the VMB system

Introduce velocity weight functions $w_1(w_2)$ for the VMB system without (with) cutoff: $w_i(t,v) := \exp\left(\frac{\langle v\rangle^i}{8RT_c \ln(\mathrm{e}+t)}\right)$.

(i) Estimate of the nonlinear collision operator (without cutoff):

$$
\begin{aligned}
&\left|\left\langle w_1^2 \partial_\beta^\alpha \Gamma(g_1, g_2), \partial_\beta^\alpha g_3 \right\rangle\right| \\
\lesssim\ & \sum \left\{ \left|w_1 \partial_{\beta_1}^{\alpha_1} g_1\right|_{s+\gamma/2} \left|\partial_{\beta_2}^{\alpha_2} g_2\right|_{\mathbf{D}}, \left|\partial_{\beta_2}^{\alpha_2} g_2\right|_{s+\gamma/2} \left|w_1 \partial_{\beta_1}^{\alpha_1} g_1\right|_{\mathbf{D}} \right\} \left|w_1 \partial_\beta^\alpha g_3\right|_{\mathbf{D}} \\
&+ \min\left\{ \left|w_1 \partial_{\beta_1}^{\alpha_1} g_1\right|_{L^2} \left|\partial_{\beta_2}^{\alpha_2} g_2\right|_{s+\gamma/2} + \left|\partial_{\beta_2}^{\alpha_2} g_2\right|_{L^2} \left|w_1 \partial_{\beta_1}^{\alpha_1} g_1\right|_{s+\gamma/2} \right\} \left|w_1 \partial_\beta^\alpha g_3\right|_{\mathbf{D}} \\
&+ \sum \left|w_1 \partial_{\beta_1}^{\alpha_1} g_1\right|_{s+\gamma/2} \left|e^{\frac{\langle v\rangle}{8RT_c \ln(\mathrm{e}+t)}} \partial_{\beta_2}^{\alpha_2} g_2\right|_{L^2} \left|w_1 \partial_\beta^\alpha g_3\right|_{\mathbf{D}} \\
&+ \sum \left|w_1 \partial_{\beta_1}^{\alpha_1} g_1\right|_{1/2+\gamma/2} \left|\mu^{\frac{\lambda_0}{128}} \partial_{\beta_2}^{\alpha_2} g_2\right|_{L^2} \left|w \partial_\beta^\alpha g_3\right|_{1/2+\gamma/2} \\
&+ \sum [\ln(1+t)]^{-1} \left|w_1 \partial_{\beta_1}^{\alpha_1} g_1\right|_{1+\gamma/2} \left|\mu^{\frac{\lambda_0}{128}} \partial_{\beta_2}^{\alpha_2} g_2\right|_{L^2} \left|w_1 \partial_\beta^\alpha g_3\right|_{s+\gamma/2} \\
&+ \sum [\ln(1+t)]^{-2} \left|w_1 \partial_{\beta_1}^{\alpha_1} g_1\right|_{1+\gamma/2} \left|\mu^{\frac{\lambda_0}{128}} \partial_{\beta_2}^{\alpha_2} g_2\right|_{L^2} \left|w_1 \partial_\beta^\alpha g_3\right|_{1+\gamma/2},
\end{aligned}
$$

(ii) Estimate of the transport term $v \cdot \nabla_x h^{\epsilon}$:

Noting that for $|\beta| \geq 1$,

$$|\langle \partial_\beta^\alpha (v \cdot \nabla_x h^\varepsilon), w_i^2 \partial_\beta^\alpha h^\varepsilon \rangle| \lesssim \left\| w_i \nabla_x^{|\alpha|+1} \nabla_v^{|\beta|-1} h^\varepsilon \right\| \left\| w_i \partial_\beta^\alpha h^\varepsilon \right\|,$$

we can obtain

$$\begin{aligned} & |\langle \partial_\beta^\alpha (v \cdot \nabla_x h^\varepsilon), w_i^2 \partial_\beta^\alpha h^\varepsilon \rangle| \\ \lesssim \quad & \varepsilon^{\frac{1}{6}} \Big(\frac{1}{\varepsilon} \left\| w_i \partial_\beta^\alpha h^\varepsilon \right\|_{\mathbf{D}}^2 + Y(t) \left\| \langle v \rangle^{\frac{i}{2}} w_i \partial_\beta^\alpha h^\varepsilon \right\|^2 \\ & + \frac{1}{\varepsilon} \left\| w_i \nabla_x^{|\alpha|+1} \nabla_v^{|\beta|-1} h^\varepsilon \right\|_{\mathbf{D}}^2 + Y(t) \left\| \langle v \rangle^{\frac{i}{2}} w_i \nabla_x^{|\alpha|+1} \nabla_v^{|\beta|-1} h^\varepsilon \right\|^2 \Big). \end{aligned}$$

(iii) Estimates of the nonlinear terms:

$$\frac{E \cdot v}{2RT_c} h^\varepsilon, \qquad \big(E + v \times B \big) \cdot \nabla_v h^\varepsilon$$

(iv) Construction of coefficients $(F_n, E_n, B_n), 1 \leq n \leq 2k-1$ (without cutoff):

For $q \in (0, \infty)$, $\ell \geq 0$,

$$\left|\partial_t^{\alpha_0}\partial_\beta^\alpha\left(\mu^{-\frac{1}{2}}F_n\right)\right| \lesssim \langle v\rangle^{-\ell+n+|\gamma+2s|/2}e^{-q\langle v\rangle}(1+t)^n. \tag{4}$$

Main difficulty to prove (4): $\frac{\mathbf{M}}{\mu} \sim e^{c_0\langle v\rangle^2}$.

Key points to show (4):

- Set $F_n = \mathbf{M}^{\frac{1}{2}}f_n = \mu^{\frac{1}{2}}h_n$
- Solve F_n via f_n, while prove (4) via h_n

Thank you for your attentions!